(ii) $\sin^2\alpha+\sin^2\beta+\sin^2\gamma=2$.

(iii) the direction cosines of $\vec{r}$ are $\frac{x}{\sqrt{x^2+y^2+z^2}},\frac{y}{\sqrt{x^2+y^2+z^2}},\frac{z}{\sqrt{x^2+y^2+z^2}}$.

(iv) l, m, n are the direction cosines of a vector if and only if $l^2+m^2+n^2=1$.

(v) any unit vector can be written as $\cos\alpha\hat{i}+\cos\beta\hat{j}+\cos\gamma\hat{k}$

Proof

(i) $\cos^2\alpha+\cos^2\beta+\cos^2\gamma=\frac{x^2}{r^2}+\frac{y^2}{r^2}+\frac{z^2}{r^2}=\frac{x^2+y^2+z^2}{r^2}=\frac{r^2}{r^2}=1$

The proofs of (ii), (iii), (iv), and (v) are left as exercise.

Example 8.5

Find a direction ratio and direction cosines of the following vectors.

(i) $3\hat{i}+4\hat{j}-6\hat{k}$, (ii) $3\hat{i}-4\hat{k}$.

Solution

(i) The direction ratios of $3\hat{i}+4\hat{j}-6\hat{k}$ are 3, 4, – 6.

The direction cosines are $\left(\frac{x}{r},\frac{y}{r},\frac{z}{r}\right)$, where $r=\sqrt{x^2+y^2+z^2}$.

Therefore, the direction cosines are $\left(\frac{3}{\sqrt{61}},\frac{4}{\sqrt{61}},\frac{-6}{\sqrt{61}}\right)$

(ii) The direction ratios of $3\hat{i}-4\hat{k}$ are 3, 0, – 4.

The direction cosines are $\left(\frac{3}{5},0,\frac{-4}{5}\right)$.

Example 8.6

(i) Find the direction cosines of a vector whose direction ratios are 2, 3, – 6.

(ii) Can a vector have direction angles $30°,45°,60°$?

(iii) Find the direction cosines of $\overrightarrow{AB}$, where A is (2, 3, 1) and B is (3, – 1, 2).

(iv) Find the direction cosines of the line joining (2, 3, 1) and (3, – 1, 2).

(v) The direction ratios of a vector are 2, 3, 6 and it's magnitude is 5. Find the vector.

Solution

(i) The direction cosines are $\left(\frac{x}{\sqrt{x^2+y^2+z^2}},\frac{y}{\sqrt{x^2+y^2+z^2}},\frac{z}{\sqrt{x^2+y^2+z^2}}.\right)$

That is, $\left(\frac{2}{7},\frac{3}{7},\frac{-6}{7}\right)$.

(ii) The condition is $\cos^2\alpha+\cos^2\beta+\cos^2\gamma=1$

Here $\alpha=30°,\beta=45°,\gamma=60°$

$\cos^2\alpha+\cos^2\beta+\cos^2\gamma=\frac{3}{4}+\frac{1}{2}+\frac{1}{4}\neq1$.

Therefore they are not direction angles of any vector.

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